A Quantitative OSINT Model for Country-Level Terrorism Financing Risk

Defensive Open-Source Intelligence and Financial Risk Analysis

Target Study Year: 2024

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This report is intended for defensive screening, academic analysis, and financial-crime risk research.

Executive Summary

This report presents a defensive open-source intelligence model for assessing country-level terrorism financing vulnerability. The model does not examine operational financing methods. Instead, it uses publicly available governance and economic indicators to create a structured risk score and compare that score against FATF grey-list status.

For the 2024 target study year, the model ranked South Sudan as the highest vulnerability country in the dataset, with a score of 81.51 out of 100.

The model uses the latest available World Bank observations up to the study year when exact country-year values are unavailable. Indicator-year fields are preserved in the output tables to make this transparent.

The study uses the most recent country-year data that can be supported consistently by the public source datasets. A 2025 or 2026 primary model year was not selected because the core World Bank governance and economic indicator datasets used in the quantitative model do not yet provide a complete and comparable 2025 or 2026 country-year panel. More recent official sources are therefore used for context and comparison, not as replacement model-year observations.

The expanded project includes supplemental conflict-exposure data, top risk-driver interpretation, regional summaries, sanctions-exposure templates, FATF mutual-evaluation templates, official AML/CFT policy context, virtual-assets and Travel Rule context, and additional visual outputs. These additions improve interpretation and future extensibility, but they do not replace the primary country-level vulnerability score.

The model should be interpreted as a screening and prioritization tool, not as a definitive determination of terrorism financing activity. Its value is in identifying countries where weak governance, political instability, conflict exposure, and economic stress may overlap with higher AML/CFT vulnerability.

Project Overview

This project develops a country-level quantitative OSINT risk model. The model is designed for defensive intelligence, financial-crime risk analysis, and academic research. It uses open-source data to estimate structural vulnerability rather than attempting to identify transactions, networks, or operational financing methods.

Research Question

Can publicly available governance and economic indicators be used to estimate country-level terrorism financing vulnerability and partially explain FATF grey-list status?

Data Sources

The model uses World Bank governance and economic indicators, combined with FATF grey-list status as a validation target. Governance indicators are drawn from the World Bank Worldwide Governance Indicators framework, including the WGI 2025 Revision / WGI 2.0 methodological context where applicable. Economic indicators are drawn from the World Bank World Development Indicators.

The study year is set to 2024 because it is the most recent year that can be supported by the project's core public indicator framework. Current FATF, Treasury, FATF, FIU, RUSI, GAO, CRS, EBA, UAE, Australia, and other official or policy sources are used as supplemental context.

Source	Indicator	Variable	Purpose
World Bank WGI	Control of Corruption	control_of_corruption	Governance integrity and corruption vulnerability.
World Bank WGI	Rule of Law	rule_of_law	Legal and institutional enforcement strength.
World Bank WGI	Political Stability	political_stability	Political instability and violence risk.
World Bank WDI	GDP per Capita	gdp_per_capita	Economic development and income vulnerability.

World Bank WDI	Inflation	inflation	Macroeconomic stress.
World Bank WDI	Unemployment	unemployment	Labor-market stress.
FATF	Grey-list status	fatf_grey_list	Validation target for model comparison.
UCDP	Organized violence country-year data	conflict_deaths_total	Supplemental conflict-exposure context.
World Bank country metadata	Region and income metadata	world_bank_region	Regional grouping and summary analysis.
Manual templates	Sanctions and FATF mutual-evaluation inputs	supplemental templates	Future structured AML/CFT enhancement layers.

Model Inputs and Analytical Role

The model separates inputs into three categories. First, core quantitative inputs are used directly in the baseline score: governance indicators, political stability, GDP per capita, inflation, and unemployment. Second, FATF grey-list status is used as a comparison and validation target, not as a direct scoring input. Third, conflict data, regional summaries, current FATF monitoring status, sanctions templates, mutual-evaluation templates, and policy sources are supplemental interpretation layers.

This separation is important for model governance. The baseline score remains transparent and reproducible because it is calculated from a small set of public country-level indicators. Supplemental layers help analysts interpret why a country may deserve additional review, but they do not change the score unless they are explicitly added in a future model version.

Methodology

The project uses a country-level risk scoring approach. The model pulls World Bank indicators through the public API, cleans the country-level dataset, removes regional aggregates, creates risk-oriented features, normalizes each feature to a common 0-to-1 scale, and applies weighted scoring. When exact study-year observations are unavailable for a country or indicator, the pipeline keeps the latest available observation up to the study year and records the year used.

This latest-available rule is a deliberate data-quality choice. The study does not advance the primary model year to 2025 or 2026 because doing so would require mixing unavailable, incomplete, or non-comparable country-year observations across the core indicators.

Conflict event data, regional summaries, sanctions exposure templates, FATF mutual-evaluation templates, official policy sources, and virtual-asset regulatory context are added as supplemental layers. They are used to improve interpretation and future extensibility, not to redefine the baseline terrorism-financing vulnerability score.

Process Summary

- Collect World Bank governance and economic data for all available countries using public API endpoints.
- Select the latest available observation up to the target study year for each indicator and preserve the source year.
- Remove regional aggregates and retain modeled country-level observations with complete core risk features.
- Convert raw indicators into risk-oriented variables so higher values consistently represent higher vulnerability.
- Normalize all risk features to a common 0-to-1 scale and apply transparent baseline weights.
- Rank countries by the final 0-to-100 terrorism-financing vulnerability score.
- Fit a logistic classification model to compare the risk features against FATF grey-list status.
- Add supplemental interpretation layers including risk tiers, sensitivity analysis, conflict exposure, regional summaries, and current FATF comparison.

Risk Feature Construction

Indicator	Risk Interpretation	Model Treatment
Control of Corruption	Lower control of corruption suggests weaker public-sector integrity.	Inverted so higher values represent higher risk.
Rule of Law	Lower rule of law suggests weaker legal and	Inverted so higher values represent higher risk.

	institutional enforcement.	
Political Stability	Lower political stability suggests instability and violence risk.	Inverted so higher values represent higher risk.
GDP per Capita	Lower income may indicate greater economic vulnerability.	Log transformed and inverted.
Inflation	Higher inflation may indicate macroeconomic stress.	Higher values treated as higher risk.
Unemployment	Higher unemployment may indicate economic stress.	Higher values treated as higher risk.

Risk Score Weights

- Corruption risk: 25%
- Rule of law risk: 20%
- Political instability risk: 25%
- Low income risk: 10%
- Inflation risk: 10%
- Unemployment risk: 10%

Indicator Year Coverage

The table below summarizes the year coverage retained in the final rankings file. This supports transparency when the latest available observation up to the study year is used.

Indicator Year Field	Most Common Year	Exact Study-Year Rows	Countries
control_of_corruption_year	2023	0	173
rule_of_law_year	2023	0	173
political_stability_year	2023	0	173
gdp_per_capita_year	2024	166	173
inflation_year	2024	163	173
unemployment_year	2024	169	173

Top Country Risk Rankings

The table below shows the top 20 countries by terrorism-financing vulnerability score. The score is a structural vulnerability indicator, not evidence of terrorist-financing activity.

Rank	Country	Score	Grey List	Pred. Prob.
1	South Sudan	81.51	Yes	86.62%
2	Syrian Arab Republic	80.54	Yes	86.35%
3	Yemen, Rep.	79.51	Yes	78.58%
4	Sudan	78.76	No	75.96%
5	Venezuela, RB	75.96	Yes	84.90%
6	Afghanistan	74.44	No	70.50%
7	Libya	72.86	No	82.26%
8	Central African Republic	71.99	No	80.94%
9	Iraq	71.26	No	82.35%
10	Congo, Dem. Rep.	70.75	Yes	83.10%
11	Myanmar	68.09	No	82.81%
12	Haiti	67.40	Yes	71.28%
13	Mali	66.69	No	66.52%
14	Burundi	65.24	No	73.86%
15	Chad	64.96	No	81.20%
16	Lebanon	64.57	Yes	68.56%
17	Zimbabwe	64.10	No	66.65%
18	Iran, Islamic Rep.	63.22	No	72.09%
19	Pakistan	62.77	No	62.73%
20	Nigeria	62.71	No	63.44%

How to Read the Rankings

The risk rank orders countries from highest to lowest structural vulnerability based on the weighted score. A higher score means the country has a stronger combination of weak governance, political instability, lower income conditions, inflation stress, or unemployment stress under the model assumptions.

The Grey List column shows whether the country is coded as a FATF grey-list jurisdiction in the model target variable. The predicted probability is the model's estimated probability of grey-list status based on the selected risk features. It is useful for screening and comparison, but it is not a finding of terrorist-financing activity.

FATF Grey-List Background

The Financial Action Task Force, or FATF, is the global standard-setting body for anti-money laundering, countering the financing of terrorism, and countering proliferation financing. FATF publishes public lists of jurisdictions with strategic weaknesses in their AML/CFT/CPF regimes.

The term 'grey list' commonly refers to FATF's list of jurisdictions under increased monitoring. These jurisdictions are working with FATF to address strategic deficiencies in their regimes for countering money laundering, terrorist financing, and proliferation financing. When a jurisdiction is placed under increased monitoring, it has committed to resolve identified deficiencies within agreed timeframes and is subject to additional monitoring.

Grey-list status does not mean that terrorist financing is occurring in a jurisdiction, and it should not be interpreted as evidence of terrorist-financing activity. It is a policy and regulatory monitoring status that reflects strategic AML/CFT/CPF deficiencies and a formal commitment to remediate them.

In this project, FATF grey-list status is used as a comparison and validation target. The model asks whether broad public indicators such as corruption control, rule of law, political stability, income, inflation, and unemployment help explain which countries appear under increased monitoring. FATF status is not used as a direct scoring input in the baseline vulnerability score.

This distinction is important. A country may receive a high model score because it has elevated structural vulnerability but may not be grey-listed by FATF. Conversely, a country may be grey-listed because of specific legal, supervisory, enforcement, beneficial ownership, financial-intelligence, or institutional deficiencies that are not fully captured by broad country-level public indicators.

Modeling Results

A logistic regression model was used to evaluate whether the scaled risk indicators help explain FATF grey-list status. Because FATF grey-listed countries represent a small minority of the dataset, the classification model used balanced class weights.

The model input matrix consists of the six scaled risk features used in the baseline score. The target variable is the binary FATF grey-list flag. The model output is a predicted probability that allows the ranked vulnerability score to be compared with observed FATF monitoring status. This comparison is useful for validation and prioritization, but it should not be interpreted as a standalone compliance conclusion.

Top Predicted Grey-List Probabilities

Country	Risk Score	Actual Grey List	Pred. Probability
South Sudan	81.51	Yes	86.62%
Syrian Arab Republic	80.54	Yes	86.35%
Venezuela, RB	75.96	Yes	84.90%
Congo, Dem. Rep.	70.75	Yes	83.10%
Myanmar	68.09	No	82.81%
Iraq	71.26	No	82.35%

Libya	72.86	No	82.26%
Russian Federation	55.56	No	82.11%
Equatorial Guinea	56.31	No	81.91%
Chad	64.96	No	81.20%

Interpretation

The model should be interpreted as moderately predictive rather than definitive. FATF grey-list status is influenced by detailed AML/CFT evaluations, legal reforms, institutional capacity, enforcement effectiveness, and policy commitments. These factors are only partially captured by broad governance and economic indicators.

The strongest intelligence value of the model is its use as an early-stage screening tool. It can help analysts identify countries where governance weakness, political instability, and economic stress overlap, creating a higher-risk environment that may warrant closer review.

This model aligns with official public-sector guidance emphasizing that terrorist and illicit financing risk is best understood through multiple indicators and the totality of circumstances.

Practical Use Case for AML/CFT Screening

The model can support early-stage geographic risk screening for financial crime, AML/CFT, and risk analytics teams. It is not intended to make final risk decisions, but it can help prioritize countries for deeper qualitative review and support structured discussion of governance, instability, conflict, and economic stress indicators.

- Country-risk screening and geographic AML/CFT review.
- Prioritization of countries for deeper qualitative analysis.
- Comparison of internal risk views against public data indicators.
- Support for risk committee discussion and model documentation.
- Supplemental context for FATF monitoring, conflict exposure, and regional risk analysis.

Enhanced Analysis

This section expands the model interpretation beyond the country ranking. It translates numerical scores into risk tiers, summarizes model classification performance, reviews false positives and false negatives, and tests whether the rankings remain stable under alternative weighting assumptions.

Risk Tier Summary

Risk Tier	Countries	Avg. Score	Avg. Probability	FATF Count
Very High	10	75.76	81.16%	5
High	11	64.54	71.00%	3
Moderate	79	49.26	50.07%	8
Low	48	30.13	20.81%	3
Very Low	25	13.24	5.68%	0

The tier results make the numerical score easier to communicate. Countries in the Very High and High tiers have the strongest overlap of governance weakness, political instability, and economic stress.

Model Performance Summary

Metric	Value
Accuracy	0.705
Precision, Grey-List Class	0.167
Recall, Grey-List Class	0.400
F1 Score, Grey-List Class	0.235
ROC AUC	0.692
Test Set Size	44

Grey-List Cases in Test Set	5
Non-Grey-List Cases in Test Set	39
Grey-List Cases, Total	19
Non-Grey-List Cases, Total	154

The performance metrics show screening value but limited classification certainty. This is expected because FATF grey-list status is influenced by policy, supervisory, enforcement, and institutional factors not fully captured by broad country-level open-source indicators.

False Positive and False Negative Review

The classification review uses a 50% predicted-probability threshold. False positives are not evidence of terrorist financing activity. They are countries where the model detects elevated structural vulnerability even though the country is not coded as FATF grey-listed. False negatives are FATF grey-listed countries that the model assigns a probability below the threshold.

False positives identified: 47 countries. The table shows the highest-probability false positives.

Country	Rank	Score	Pred. Prob.
Myanmar	11	68.09	82.81%
Iraq	9	71.26	82.35%
Libya	7	72.86	82.26%
Russian Federation	35	55.56	82.11%
Equatorial Guinea	34	56.31	81.91%
Chad	15	64.96	81.20%
Central African Republic	8	71.99	80.94%
Nicaragua	37	54.76	78.30%

False negatives identified: 7 countries. The table shows FATF grey-listed countries below the 50% model threshold.

Country	Rank	Score	Pred. Prob.
Namibia	109	37.16	10.61%
Kuwait	124	29.84	30.02%
Nepal	63	48.99	36.46%
Bulgaria	112	35.45	38.31%
Kenya	48	52.04	46.40%
Cote d'Ivoire	71	46.87	46.61%
Algeria	58	50.78	49.64%

Analytically, false positives can be treated as candidates for further qualitative review, while false negatives show where FATF monitoring may reflect detailed institutional, legal, or supervisory weaknesses not fully represented in the selected public indicators.

Sensitivity Analysis

Scenario	Top-20 Overlap	Baseline Count	Overlap %
Baseline	20	20	100.00%
Equal Weight	18	20	90.00%
Governance Heavy	20	20	100.00%
Political Instability Heavy	19	20	95.00%
Economic Stress Heavy	18	20	90.00%

The sensitivity results indicate whether the top country group remains reasonably stable across weighting assumptions. This strengthens the usefulness of the model as a transparent screening tool while still showing that exact rank order depends on how governance, instability, and economic stress are weighted.

Top Risk Drivers

This section improves model interpretability by identifying the largest risk-score contributors for the highest-ranked countries. The driver labels explain which normalized risk factors most strongly contributed to each country's vulnerability score.

Rank	Country	Score	Driver 1	Driver 2	Driver 3
1	South Sudan	81.51	Corruption risk	Political instability risk	Rule of law risk
2	Syrian Arab Republic	80.54	Political instability risk	Corruption risk	Rule of law risk
3	Yemen, Rep.	79.51	Political instability risk	Corruption risk	Rule of law risk
4	Sudan	78.76	Political instability risk	Corruption risk	Rule of law risk
5	Venezuela, RB	75.96	Corruption risk	Rule of law risk	Political instability risk
6	Afghanistan	74.44	Political instability risk	Corruption risk	Rule of law risk
7	Libya	72.86	Corruption risk	Political instability risk	Rule of law risk
8	Central African Republic	71.99	Political instability risk	Corruption risk	Rule of law risk
9	Iraq	71.26	Political instability risk	Corruption risk	Rule of law risk
10	Congo, Dem. Rep.	70.75	Corruption risk	Political instability risk	Rule of law risk

Risk drivers should be read as model explanations, not as independent proof of terrorist-financing activity. They show why the scoring framework placed a country high in the ranking.

Conflict Event Data Supplement

This supplemental section adds country-level organized-violence context using the UCDP country-year organized-violence dataset. Conflict exposure is relevant to terrorism-financing vulnerability because conflict, instability, and weak state capacity can affect AML/CFT risk environments. The conflict supplement does not replace the main vulnerability score and is not evidence of terrorist-financing activity.

Within the modeled dataset, 169 countries were matched to UCDP country-year records. The supplement identifies 56 countries with recorded organized-violence deaths and 150,567 total recorded organized-violence deaths in the matched supplement.

Top Countries by Recorded Organized-Violence Deaths

Country	Rank	Risk Score	UCDP Year	Deaths	Conflict Events	Category
Ukraine	28	57.12	2024	68,116	2	High recorded conflict exposure
Israel	113	34.97	2024	21,476	6	High recorded conflict exposure
Mexico	59	50.22	2024	11,022	24	High recorded conflict exposure
Sudan	4	78.76	2024	7,103	8	High recorded conflict exposure
Congo, Dem. Rep.	10	70.75	2024	5,409	29	High recorded conflict exposure
Lebanon	16	64.57	2024	4,339	5	High recorded conflict exposure
Burkina Faso	26	57.77	2024	4,251	6	High recorded conflict exposure
Ethiopia	23	59.11	2024	4,072	14	High recorded conflict exposure
Syrian Arab	2	80.54	2024	3,616	15	High recorded

Republic						conflict exposure
Nigeria	20	62.71	2024	3,170	35	High recorded conflict exposure
Myanmar	11	68.09	2024	2,896	16	High recorded conflict exposure
Mali	13	66.69	2024	1,901	15	High recorded conflict exposure

Large conflict-death counts are treated as supplemental instability context. They should be combined with governance, AML/CFT, sanctions, FATF, and qualitative context before any country-level risk conclusion is drawn.

Additional Risk Factor Analysis

This section summarizes supplemental risk-factor outputs added to make the project more extensible. Regional summaries provide geographic structure, while the sanctions and FATF mutual-evaluation templates create input files for future structured enhancements. These outputs are supplemental and do not replace the baseline model score.

Regional Risk Summary

Region	Country Count	Average Risk Score	Median Risk Score	FATF Grey List Count	Current FATF Monitoring Count	Highest Risk Country
Sub-Saharan Africa	45	53.33	53.89	7	7	South Sudan
Middle East, North Africa, Afghanistan & Pakistan	22	49.52	49.75	5	5	Syrian Arab Republic
Latin America & Caribbean	28	42.27	42.72	3	3	Venezuela, RB
South Asia	6	41.60	44.53	1	1	Bangladesh
East Asia & Pacific	23	33.37	38.76	2	3	Myanmar
Europe & Central Asia	47	30.03	28.35	1	1	Tajikistan
North America	2	17.49	17.49	0	0	United States

Sanctions and FATF Mutual Evaluation Templates

- Sanctions exposure supplement created with 173 country rows.
- FATF mutual-evaluation supplement created with 173 country rows.

These files are intentionally created as country-level input templates. The 173 rows indicate that every modeled country has a placeholder record that can be populated later with structured sanctions, FATF technical-compliance, or FATF effectiveness data. The current template rows should not be interpreted as evidence of sanctions exposure, FATF deficiencies, or terrorism-financing activity.

This design keeps the baseline model transparent. The current vulnerability score continues to rely only on the core governance and economic indicators. Sanctions exposure and FATF mutual-evaluation data can be added later as separate supplemental layers once reliable, country-comparable source values are collected.

Additional Risk Factor Output Summary

Factor	Status	Records	Output
Regional risk groupings	Created from World Bank country metadata	7	outputs/tables/regional_risk_summary.csv
Regional conflict context	Created if conflict supplement exists; defaults to zero otherwise	56	outputs/tables/regional_risk_summary.csv
Sanctions exposure indicators	Template created; supplement updates when template is populated	0	outputs/tables/sanctions_exposure_supplement.csv

FATF mutual evaluation indicators	Template created; supplement updates when template is populated	0	outputs/tables/fatf_mutual_evaluation_supplement.csv
Current FATF visualization	Chart created if current FATF comparison file exists	1	outputs/charts/current_fatf_status_comparison.png

Current Data Supplement and Forward-Looking Risk Context

This section compares the model outputs with the current FATF increased-monitoring status configured in the project and current economic indicators collected as a supplement. The supplement does not replace the main model; it provides an additional current-status comparison layer.

This distinction matters because FATF monitoring status and official risk guidance can be more current than the structured country-year indicator datasets used for model scoring. The project therefore separates the quantitative model year from forward-looking context.

In this run, the current economic supplement uses the same target year as the baseline economic indicators. As a result, the economic change fields may show 0.00 when the latest available current value is the same observation used in the baseline model. The main analytical value of this section is therefore the current FATF status comparison and the ability to refresh this layer in future model runs.

Within the modeled dataset, 20 countries are coded as currently under FATF increased monitoring in the project configuration. The current-status comparison flags 1 country where the model FATF target and current FATF status differ. The model's 50% probability signal matches current FATF status for 118 countries out of 173.

Current FATF Monitored Jurisdictions by Model Probability

Country	Rank	Score	Pred. Prob.	Model FATF	Current FATF
South Sudan	1	81.51	86.62%	Yes	Yes
Syrian Arab Republic	2	80.54	86.35%	Yes	Yes
Venezuela, RB	5	75.96	84.90%	Yes	Yes
Congo, Dem. Rep.	10	70.75	83.10%	Yes	Yes
Yemen, Rep.	3	79.51	78.58%	Yes	Yes
Bolivia	54	51.27	73.66%	Yes	Yes
Cameroon	21	60.21	71.90%	Yes	Yes
Haiti	12	67.40	71.28%	Yes	Yes
Lebanon	16	64.57	68.56%	Yes	Yes
Lao PDR	79	44.62	63.15%	Yes	Yes
Papua New Guinea	66	48.16	54.61%	Yes	Yes
Angola	39	54.31	51.49%	Yes	Yes

Economic Indicator Change Summary

Measure	Mean	Median	Available Rows
GDP per Capita Change %	0.00	0.00	173
Inflation Change, Points	0.00	0.00	173
Unemployment Change, Points	0.00	0.00	173

These change measures are supplemental. In this run, the baseline model and current economic supplement rely on the same latest-available economic observations, so the reported changes are 0.00. In future runs, this table can help identify whether economic conditions have shifted relative to the model's main indicator values. These changes should not be interpreted as direct evidence of terrorist-financing activity.

Official Terrorism Designation Context by Region

A regional designation context layer could add interpretive value to the project, but it should be framed carefully. The report should avoid describing a broad list of 'confirmed terrorist groups.' A more accurate description is official terrorism designation context, because designations depend on the legal authority, jurisdiction, date, and list used.

This section is therefore treated as supplemental context only. Future versions could summarize the number of officially designated entities by region using official sources such as the UN Security Council ISIL/Da'esh and Al-Qaida sanctions list, the U.S. Department of State Foreign Terrorist Organizations list, and comparable official national or multilateral designation lists.

The safest use in this model is a high-level regional count or source-reference table. Detailed group profiles, operational descriptions, tactical details, or financing-method descriptions are outside the scope of this defensive country-level screening model.

South Africa Case Study: FATF Grey-List Remediation and FIU Context

South Africa provides a useful country-level example of how FATF grey-list status can reflect institutional and supervisory deficiencies rather than direct evidence of terrorist-financing activity. The Financial Intelligence Centre Annual Report 2024/25 states that South Africa was grey listed by FATF in February 2023 and that, by the end of the reporting year, the country had addressed 20 of 22 FATF action-plan items.

The report also notes that FATF announced in June 2025 that South Africa had substantially completed all 22 action-plan items, subject to verification. This supports the study's treatment of FATF status as a policy and regulatory benchmark rather than a direct terrorism-financing activity measure.

The South Africa case also shows why country-level structural indicators should be interpreted with institutional context. A jurisdiction's FATF status may change as legal reforms, supervisory improvements, beneficial ownership controls, enforcement effectiveness, and financial intelligence capabilities develop over time.

U.S. Treasury 2024 National Terrorist Financing Risk Assessment Context

The U.S. Treasury 2024 National Terrorist Financing Risk Assessment provides official government context for terrorist-financing threats, vulnerabilities, financial channels, and risk-assessment methodology. It is useful for this study because it reinforces the need to interpret country-level vulnerability alongside more detailed financial-sector, enforcement, and typology information.

The assessment supports the report's limitations because it discusses multiple channels and vulnerability areas that the baseline model does not directly measure, including money services businesses, person-to-person payments, unregistered money transmission, cash, banks, virtual assets, non-profit organizations, crowdfunding, and online fundraising.

This context strengthens the study's conclusion that the OSINT model is a screening framework, not a complete terrorist-financing detection system. It can help prioritize geographic review, but higher-confidence assessment requires transaction-level indicators, institutional controls, law enforcement information, financial intelligence, and qualitative analysis.

U.S. Treasury 2024 Illicit Finance Strategy Context

The U.S. Treasury 2024 National Strategy for Combating Terrorist and Other Illicit Financing provides broader strategic context for this project. It describes the U.S. government's approach to denying illicit actors access to the funds and resources needed to carry out harmful activities or profit from crimes.

This strategy fits the model's defensive framing because it emphasizes risk-based prioritization, AML/CFT modernization, transparency, supervision, enforcement, and disruption of illicit-finance networks. The model's value is not final classification; it is a structured way to prioritize geographic risk review using transparent public indicators.

The strategy also supports future expansion into legal, supervisory, transparency, sanctions, and enforcement indicators. Those indicators should remain separate supplemental layers until reliable and country-comparable data are available.

Virtual Assets Context: Privacy Coins and Illicit-Finance Risk

Qualitative research on privacy coins and illicit-finance risk provides useful context for the virtual-assets limitations of this country-level model. Some reporting suggests that certain illicit actors or support networks may show interest in privacy-focused cryptocurrency because privacy features can complicate tracing compared with more transparent virtual assets.

This context should not be treated as a model variable in the current version. The baseline model does not observe wallets, transactions, blockchain flows, crowdfunding campaigns, privacy-coin usage, or online fundraising appeals. It only measures country-level structural vulnerability.

Future versions could add a virtual-assets supplement using official enforcement actions, FIU reports, national risk assessments, Travel Rule implementation indicators, regulated-sector typology reports, and country-level crypto-asset supervision indicators.

UAE AML/CFT/CPF National Strategy 2024-2027 Context

The UAE AML/CFT/CPF National Strategy 2024-2027 provides a useful example of how jurisdictions can continue strengthening AML/CFT frameworks after FATF-related reforms. The strategy emphasizes risk-based compliance, supervision, national and international coordination, detection and disruption of illicit financial activity, legal and regulatory updates, and emerging risks.

This source fits the report because the UAE was removed from the FATF grey list in February 2024, while the national strategy shows continued attention to AML/CFT and counter-proliferation-financing capabilities. It illustrates why FATF status should be interpreted dynamically rather than as a static measure of country risk.

The UAE context also supports future data layers related to post-grey-list reform, supervision, emerging risks, virtual assets, trade-based money laundering, cyber-enabled financial crime, and national coordination mechanisms.

Australia AML/CTF Amendment Act 2024 Context

Australia's AML/CTF Amendment Act 2024 provides useful legal-reform context for the model's limitations and future improvements. The reform expands AML/CTF coverage, modernizes virtual-asset and payments regulation, and simplifies parts of the regime to support more flexible financial-crime prevention.

The reform is relevant because the baseline model does not directly measure domestic statutory coverage, regulated-sector scope, customer due diligence requirements, value-transfer reporting, virtual-asset service coverage, or supervisory powers. These legal and supervisory factors can affect FATF outcomes and AML/CFT effectiveness.

Future versions of the project could add a legal and regulatory supplement that tracks whether jurisdictions cover real estate professionals, precious metals and stones dealers, lawyers, conveyancers, accountants, trust and company service providers, virtual-asset services, and other high-risk sectors.

European Banking Authority Travel Rule Guidelines Context

The European Banking Authority's 2024 Travel Rule Guidelines provide useful regulatory context for the virtual-asset and funds-transfer limitations of this country-level model. The guidelines address information requirements for transfers of funds and certain crypto-asset transfers under Regulation (EU) 2023/1113.

This strengthens the study's methodological distinction between structural country risk and operational AML/CFT controls. A country may have elevated structural vulnerability, but transaction-level detection still depends on regulated institutions, crypto-asset service providers, payment service providers, supervisory expectations, and information-sharing rules.

For future development, the Travel Rule guidance could support a separate virtual-assets and payments compliance supplement. That supplement could track whether jurisdictions have implemented transfer-originator and beneficiary-information rules, whether crypto-asset service providers are covered by AML/CFT obligations, and whether supervisory frameworks address missing or incomplete transfer information.

Congressional Research Service AML/CFT Policy Overview Context

The Congressional Research Service overview of U.S. efforts to combat money laundering, terrorist financing, and other illicit financial threats provides useful policy context for this study. CRS describes the United States as maintaining a multifaceted AML/CFT and illicit-finance policy regime involving legal, regulatory, enforcement, and diplomatic tools.

The CRS overview is especially useful because it explains that terrorist financing involves fundraising through both licit and illicit means and financially sustaining terrorist groups. It also highlights that illicit financial threats include money laundering, proliferation finance, sanctions evasion, tax evasion, and other forms of state or nonstate financial facilitation.

The report reinforces the importance of statutory, regulatory, and supervisory controls that this model does not directly measure, including the Bank Secrecy Act, USA PATRIOT Act CFT provisions, FinCEN reporting, financial-institution examinations, beneficial ownership policy, real estate and investment-adviser rulemakings, and digital-asset AML/CFT regulation.

GAO Illicit-Finance Progress Assessment Context

The U.S. Government Accountability Office's 2025 review of federal illicit-finance efforts provides useful oversight context for this project. GAO examined selected federal roles, responsibilities, strategies, and efforts related to countering illicit finance activities, including terrorism financing, money laundering, transnational organized crime, and related financial-crime threats.

GAO found that federal agencies investigate and prosecute illicit-finance activity through multiple law enforcement and interagency mechanisms. The report discusses the roles of agencies such as the FBI, DEA, ICE Homeland Security Investigations, Secret Service, IRS-Criminal Investigation, and U.S. Attorneys' Offices, as well as task-force structures such as OCDETF and the El Dorado Task Force.

The GAO report is also useful for model-governance discussion because it found that selected federal strategies and efforts to counter illicit finance did not always have clearly defined goals or regular collection and assessment of performance information. That finding supports future model-governance tracking and benchmark monitoring.

RUSI 2025 Terrorist-Financing Reassessment Context

RUSI's 2025 assessment of terrorist financing provides useful qualitative context for the limitations and future-development path of this country-level OSINT model. The paper describes terrorist financing as an increasingly complex and technology-enabled landscape, with continuing use of traditional channels such as cash, hawala, and non-profit

organization abuse alongside newer channels such as virtual assets, fintech platforms, instant payments, online crowdfunding, and online gaming.

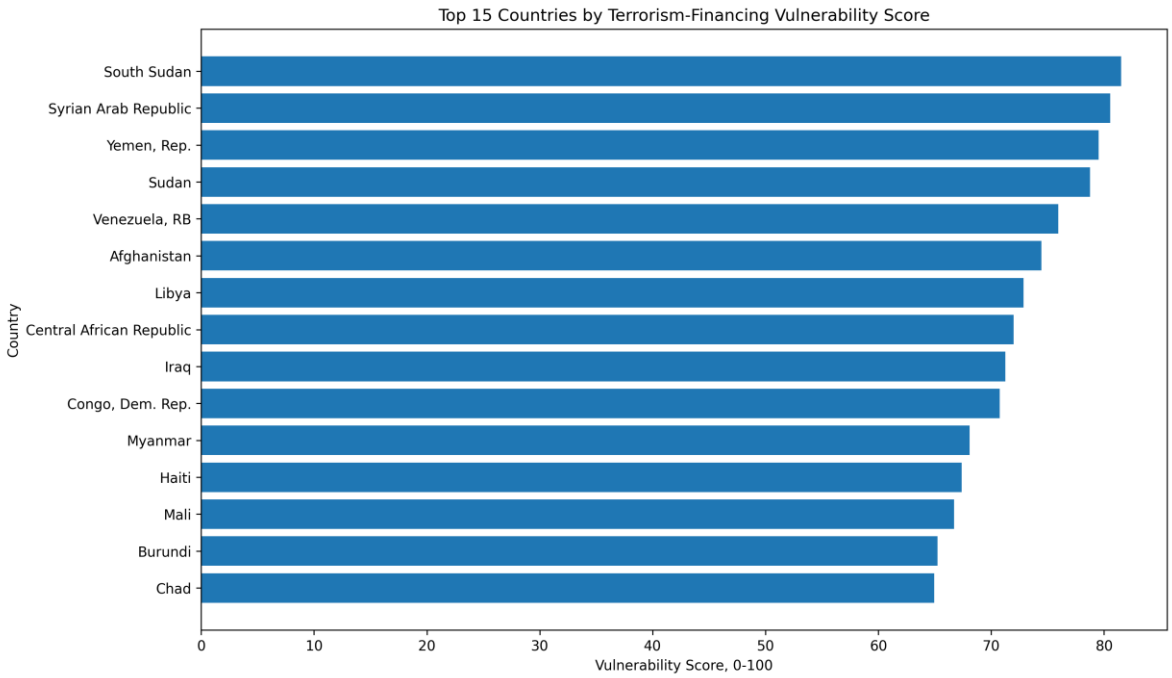
This source reinforces the difference between structural country vulnerability and operational detection. The baseline model scores broad country-level conditions such as governance weakness, political instability, and economic stress. Modern terrorist-financing detection also depends on transaction monitoring, public-private cooperation, financial intelligence sharing, virtual-asset tracing, off-chain data, platform cooperation, and investigative capacity.

Future versions could add supplemental indicators for fintech and instant-payment risk, virtual-asset and stablecoin exposure, online crowdfunding controls, online-gaming vulnerabilities, public-private partnership maturity, and supervisory guidance for emerging technology risks.

Charts and Visual Outputs

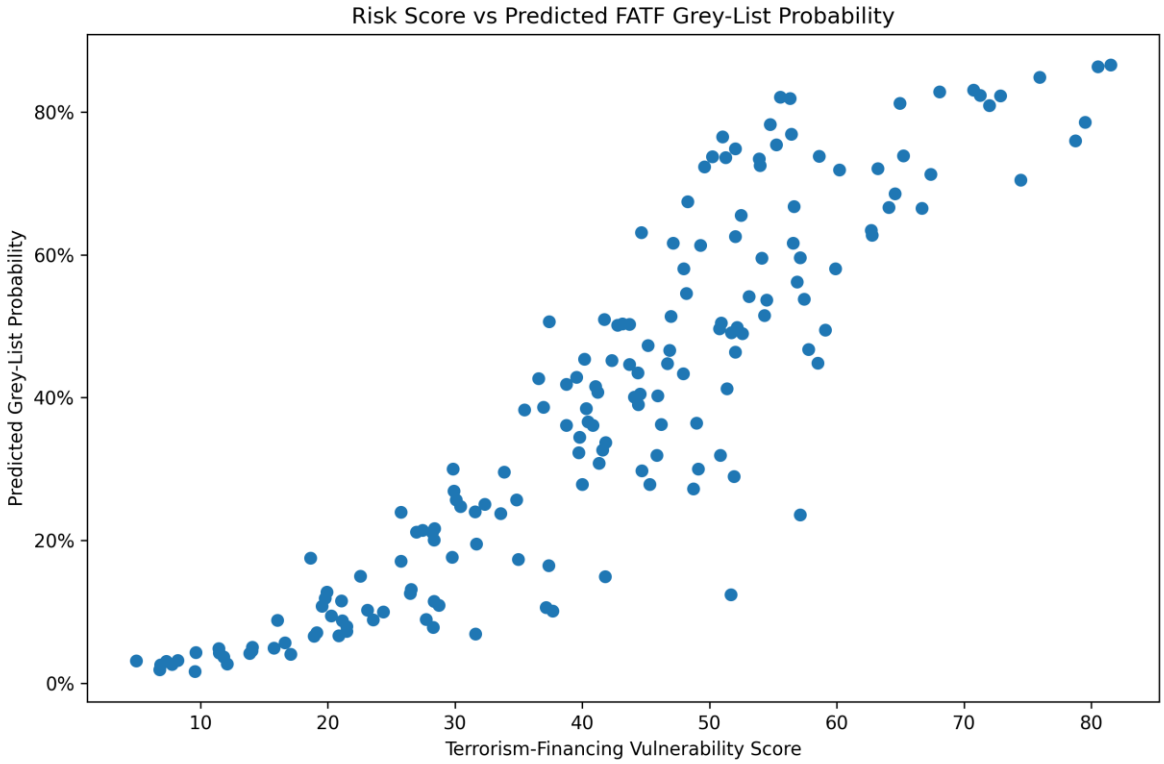
The following charts summarize the main quantitative outputs from the model. They support interpretation of the country rankings, predicted FATF grey-list probabilities, sensitivity analysis, conflict exposure, regional patterns, and current FATF comparison layer.

Figure 1. Top 15 Countries by Terrorism-Financing Vulnerability Score



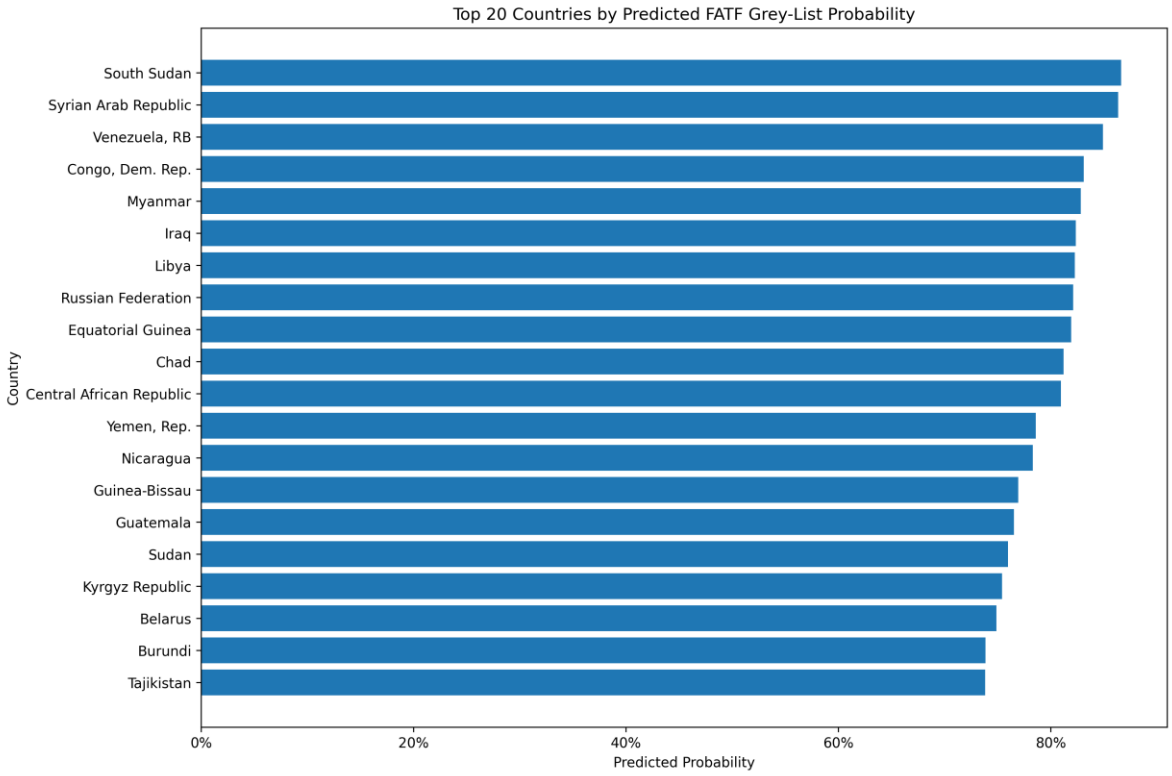
This chart shows the countries with the highest structural vulnerability scores. Higher scores reflect a stronger combination of governance weakness, political instability, lower income conditions, inflation stress, and unemployment stress under the model's weighting framework.

Figure 2. Risk Score vs Predicted FATF Grey-List Probability



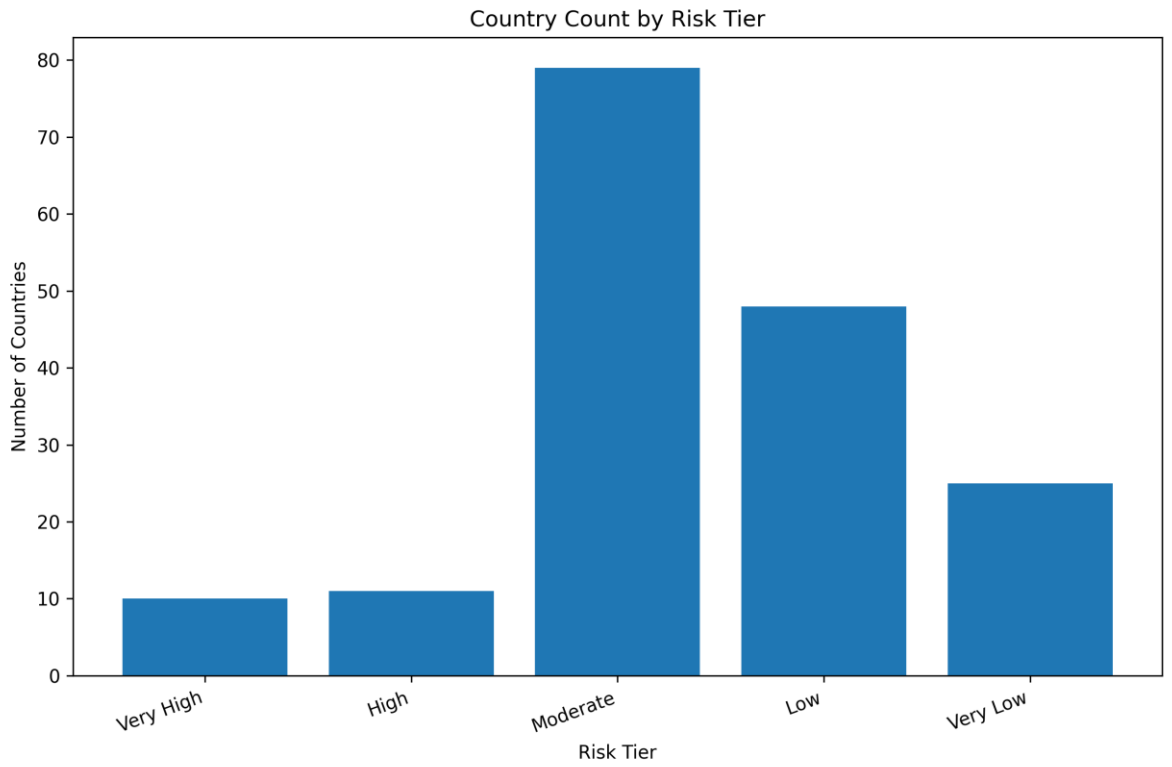
This scatter plot compares the weighted vulnerability score with the model's predicted probability of FATF grey-list status. A positive relationship suggests that the selected public indicators provide screening value, although the probability should not be treated as a definitive compliance or risk finding.

Figure 3. Top 20 Countries by Predicted FATF Grey-List Probability



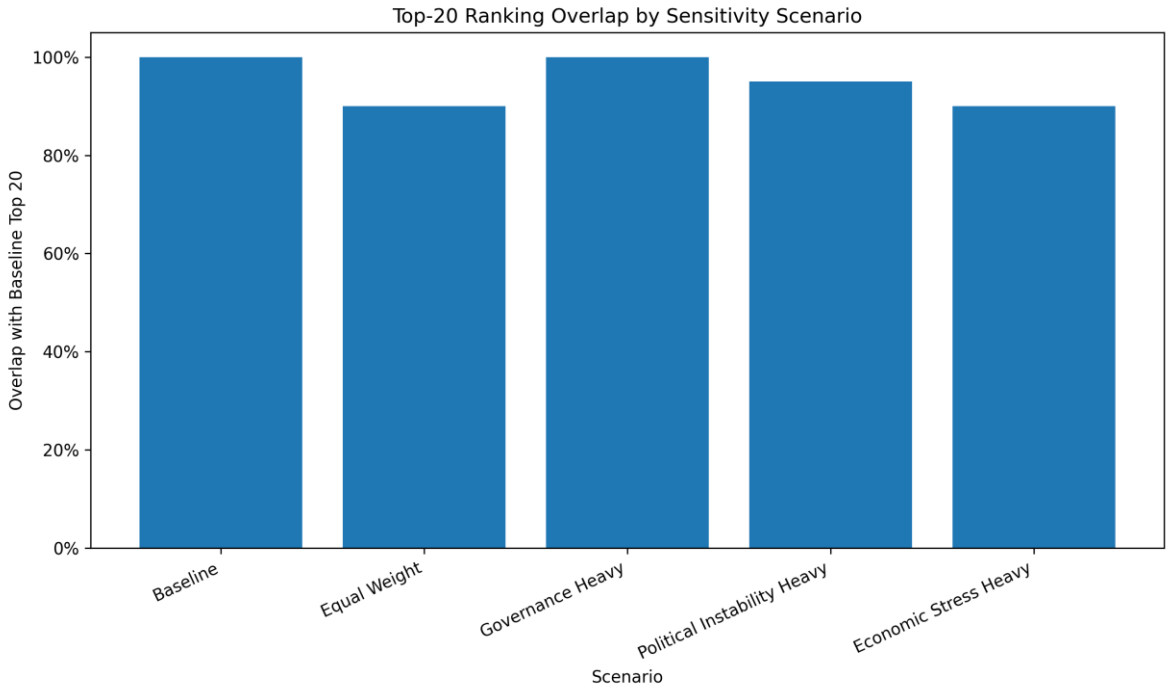
This chart ranks countries by the classification model's estimated probability of FATF grey-list status. Countries appearing here may warrant deeper qualitative review, especially when model probability is high but observed FATF status differs.

Figure 4. Risk Tier Distribution



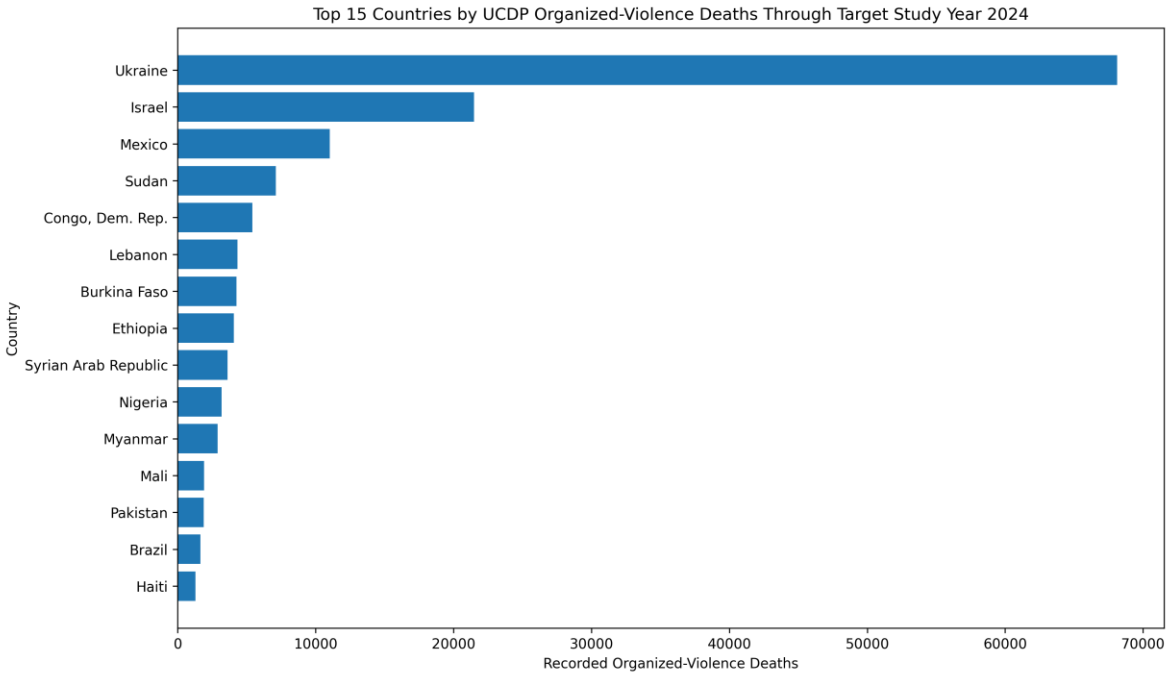
This chart shows how countries are distributed across the project's risk tiers. It translates the continuous 0-to-100 score into easier-to-communicate categories such as Very High, High, Moderate, Low, and Very Low.

Figure 5. Top-20 Sensitivity Overlap by Scenario



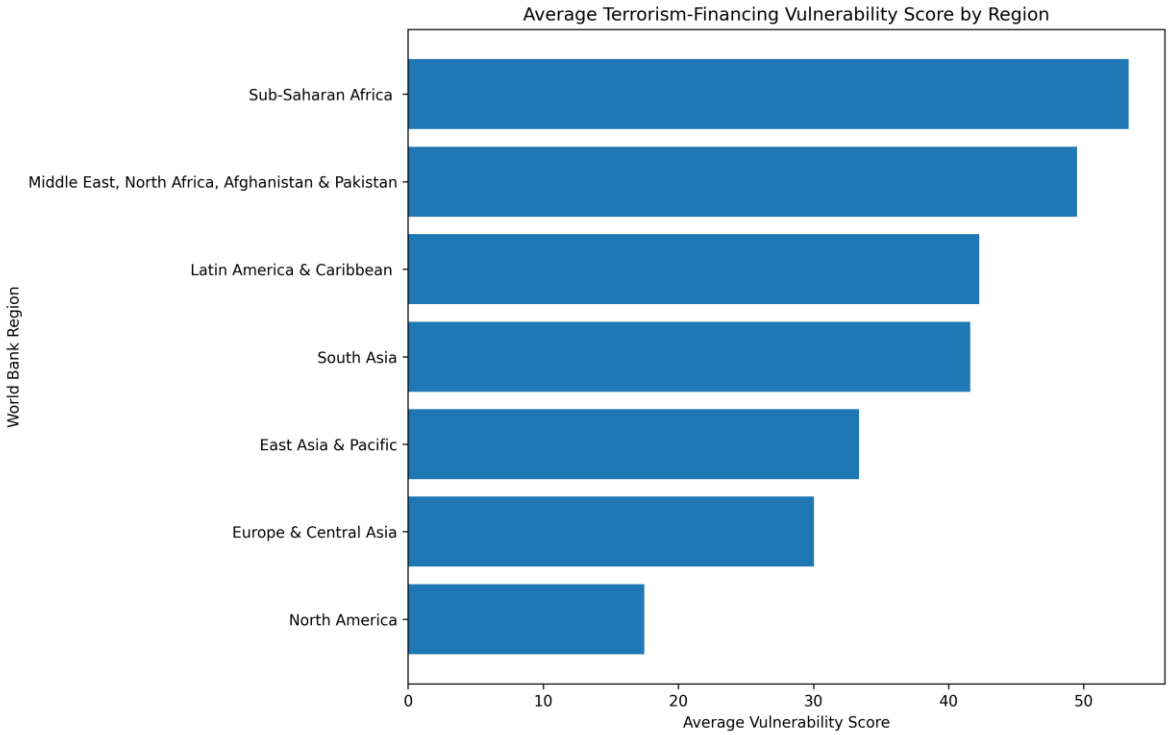
This chart tests ranking stability across alternative weighting assumptions. Higher overlap means the same countries remain in the top-risk group even when the model places more emphasis on governance, political instability, or economic stress.

Figure 6. Top 15 Countries by Recorded Organized-Violence Deaths



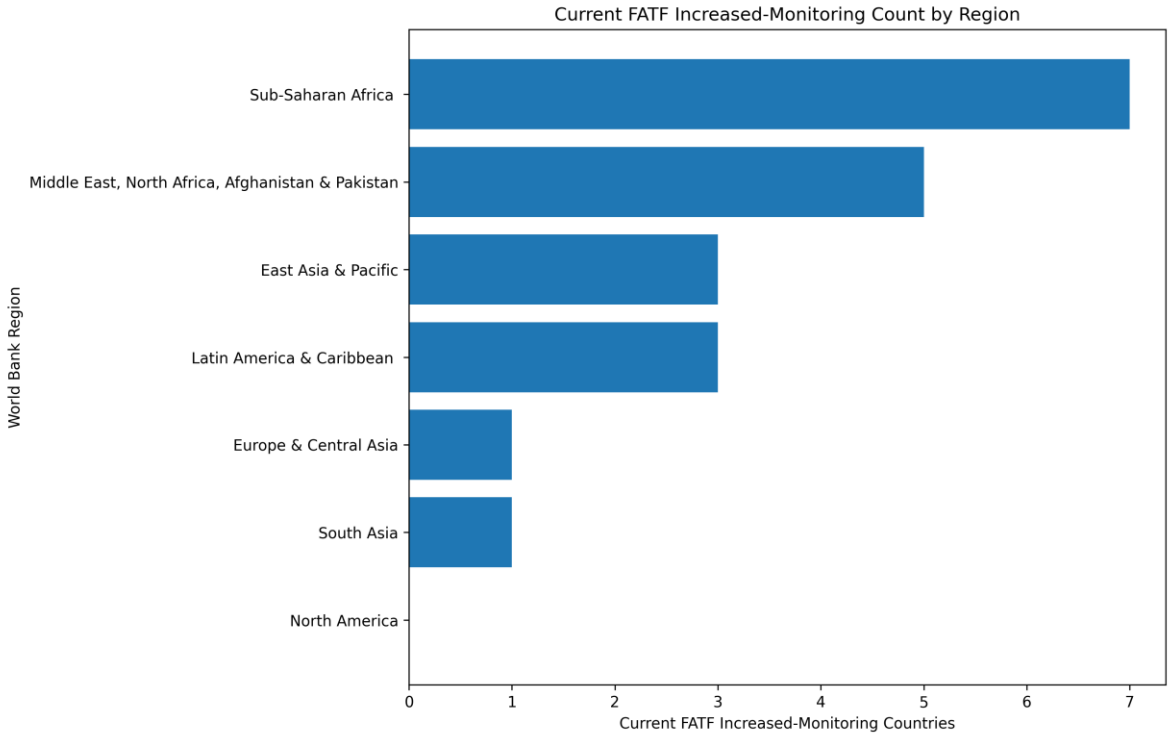
This chart summarizes supplemental conflict exposure using UCDP organized-violence data. Conflict deaths are not part of the baseline vulnerability score, but they provide useful instability context for interpreting country-level AML/CFT risk environments.

Figure 7. Regional Average Terrorism-Financing Vulnerability Score



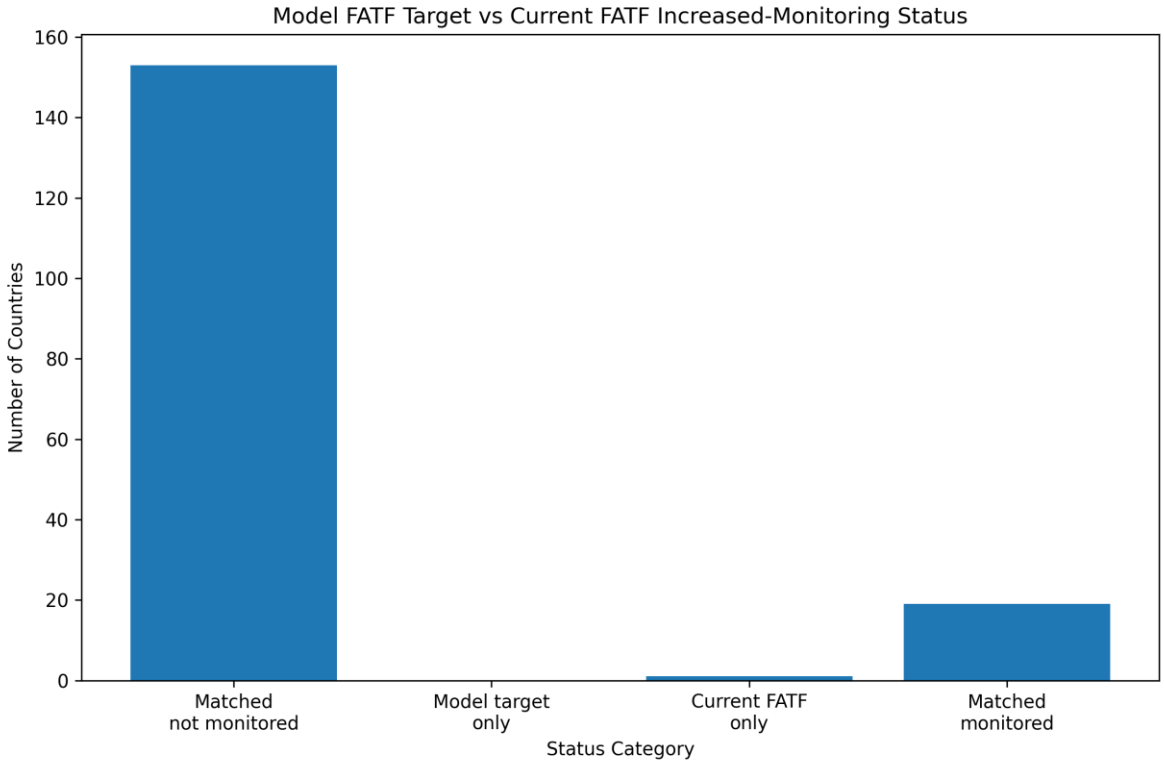
This chart compares average vulnerability scores across World Bank regional groupings. It helps identify whether elevated country-level risk is concentrated in specific regions and provides context for regional screening priorities.

Figure 8. Current FATF Increased-Monitoring Count by Region



This chart shows how current FATF increased-monitoring jurisdictions are distributed by region. It is a current-status comparison layer and should be interpreted separately from the baseline quantitative score.

Figure 9. Current FATF Status Comparison



This chart compares the model's FATF target coding with the current FATF increased-monitoring status configured in the project. It helps show where the model-year target and current-status supplement align or differ.

Correlation Analysis

The correlation matrix provides a basic view of how the model's risk indicators relate to the final vulnerability score, FATF grey-list status, and predicted grey-list probability.

Variable	Risk Score	Grey List	Pred. Prob.
Corruption Risk	0.944	0.286	0.923
Rule of Law Risk	0.962	0.307	0.940
Instability Risk	0.886	0.261	0.770
Low Income Risk	0.824	0.229	0.690
Inflation Risk	0.323	0.211	0.240
Unemployment Risk	0.222	0.098	-0.053
Risk Score	1.000	0.312	0.902
Grey List	0.312	1.000	0.292
Pred. Prob.	0.902	0.292	1.000

Limitations

- The model uses country-level indicators and does not identify individual actors, entities, or transactions.
- FATF grey-list status is a policy and regulatory outcome, not a direct measure of terrorism financing activity.
- The model does not include detailed legal, supervisory, enforcement, or financial intelligence data.
- Some countries may have missing or delayed economic data, so the pipeline records the latest available observation year used for each indicator.
- The project does not use 2025 or 2026 as the primary model year because the core public country-level indicators used for scoring are not yet available as a complete and comparable 2025 or 2026 panel.

- Conflict event data is supplemental context and is not evidence of terrorist-financing activity.
- UCDP country matching depends on country-name harmonization; unmatched countries may require a manual crosswalk.
- Sanctions exposure and FATF mutual-evaluation supplements are currently template-based unless structured source values are manually populated.
- The model is intended for defensive screening and academic analysis, not operational targeting.
- The model does not replace suspicious activity reporting, law enforcement investigations, or the multi-indicator assessment process described in official counterterrorism finance guidance.
- The model is not a substitute for institution-specific AML/CFT risk assessment processes across products, services, customers, distribution channels, and geographic locations.
- The model does not capture all current threat-specific channels discussed in national risk assessments, such as online fundraising, money services businesses, cash, digital assets, or nonprofit organization vulnerabilities.
- The model does not capture FATF's full terrorist-financing risk indicator framework, including detailed customer behavior, transaction, geographic, product/service, trade, commercial-entity, nonprofit, and organized-crime indicators.
- The model does not replace bank-level AML/CFT risk management practices, including customer due diligence, ongoing monitoring, suspicious-transaction reporting, governance arrangements, and the three lines of defense.
- The model does not directly measure virtual-asset activity, privacy-coin exposure, wallet-level behavior, crowdfunding, or online fundraising appeals.
- The model does not directly measure Travel Rule compliance, originator and beneficiary information quality, missing or incomplete transfer information, payment-service-provider controls, or crypto-asset-service-provider controls.
- The model does not directly measure domestic AML/CFT statutory coverage, beneficial ownership implementation, FinCEN-style reporting systems, supervisory examinations, enforcement actions, or congressional policy developments.
- The model does not measure interagency coordination, investigative case outcomes, prosecution results, asset forfeiture, task-force disruption metrics, or government-wide performance tracking for illicit-finance strategies.
- The model does not directly measure fintech-platform controls, instant-payment exposure, online crowdfunding activity, online-gaming financial flows, public-private partnership maturity, or cross-sector intelligence-sharing capacity.
- The current data supplement is a comparison layer and should not be treated as definitive proof of current terrorist-financing activity or absence of risk.

Conclusion

This project demonstrates how open-source data can be used to build a structured, defensive intelligence model for country-level terrorism financing vulnerability. The model combines governance weakness, political instability, and economic stress into a transparent scoring framework and compares the results against FATF grey-list status.

The expanded project adds conflict exposure, regional summaries, top risk drivers, additional charts, official AML/CFT policy context, virtual-assets context, Travel Rule context, legal and regulatory reform examples, and structured input templates for sanctions and FATF mutual-evaluation data.

The results suggest that open-source indicators can provide useful screening value, but they should be supplemented with qualitative analysis, AML/CFT evaluation reports, sanctions data, conflict event data, official risk assessments, and expert judgment for higher-confidence assessment.

Future Improvements

- Add sanctions exposure indicators with populated source data.
- Add a virtual-assets and payments compliance supplement using Travel Rule, payment-transfer, and crypto-asset transfer implementation indicators.

- Add a legal and regulatory framework supplement covering AML/CFT statutory scope, beneficial ownership transparency, supervisory coverage, enforcement activity, and digital-asset AML/CFT regulation.
- Add an emerging terrorist-financing typologies supplement covering fintech, instant payments, virtual assets, online crowdfunding, online gaming, and public-private information-sharing maturity.
- Add model-governance tracking that evaluates whether supplemental indicators improve screening performance, ranking stability, and alignment with official illicit-finance strategy benchmarks over time.
- Refine conflict event data matching with a country-name crosswalk.
- Add AML/CFT mutual evaluation data with technical compliance and effectiveness ratings.
- Expand sensitivity analysis with additional weighting scenarios.
- Refine risk-tier interpretation with regional and peer-group comparisons.
- Add additional current FATF comparison visuals over multiple FATF update cycles.
- Build an interactive Streamlit dashboard.
- Add time-series analysis across multiple years.
- Compare results across FATF grey-list updates.

References and Data Sources

The following sources were used to support the data collection, methodology, validation framework, limitations, and supplemental context for this project.

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Appendix: Logistic Regression Output

Logit Regression Results

Dep. Variable:	fatf_grey_list	No. Observations:	173
Model:	Logit	Df Residuals:	166
Method:	MLE	Df Model:	6
Date:	Fri, 15 May 2026	Pseudo R-squ.:	0.1725
Time:	09:22:18	Log-Likelihood:	-49.557
converged:	True	LL-Null:	-59.884
Covariance Type:	nonrobust	LLR p-value:	0.002115

	coef	std err	z	P>|z|	[0.025	0.975]
const	-6.1598	1.470	-4.191	0.000	-9.041	-3.279
corruption_risk_scaled	2.5986	4.094	0.635	0.526	-5.426	10.623
rule_of_law_risk_scaled	2.9941	3.996	0.749	0.454	-4.838	10.826
political_instability_risk_scaled	-0.4674	1.711	-0.273	0.785	-3.821	2.886
low_income_risk_scaled	0.6476	1.924	0.337	0.736	-3.124	4.419
inflation_risk_scaled	1.5120	1.699	0.890	0.374	-1.818	4.842
unemployment_risk_scaled	1.0645	1.419	0.750	0.453	-1.716	3.845